YOGESH KOULGEKAR

Data Scientist

Organized and skilled data scientist with more than 4 years of expertise using predictive analysis to efficiently build and connect database systems. Skilled data science with competence in statistical modeling, data mining, extraction, preparing and pre-processing data, deriving insight data, developing & deploying Machine Learning models. Interested in utilizing concrete data scientist knowledge in a fast-paced, dynamic environment to develop data gathering methods.

WORK EXPERIENCE

Carrier India | Hyderabad Nov. 2023 – Present

Infodot System Private Ltd | Bangalore Apr. 2021 – Nov. 2023

Role and Responsibilities

- Develop production ready implementations of proposed solutions across different **ML and DL algorithms**, including testing on live customer data to improve efficacy.
- Activity involving daily standup calls and task assign
- Work closely with other **functional teams** to integrate implemented systems.
- Create use cases with respect to **domain** for solving à business problem
- **Research** and **test Machine Learning** approaches for **analyzing large** scaled distributed computing applications.
- Suggest innovative and **creative concepts** and ideas that will improve the overall platform
- String understanding of recent trends around industry and adoption
- Conduct quantitative **analyses** and interpret results
- Comprehensive knowledge of **SQL, MySQL** with experience in office skills
- Working knowledge of **Optimization Techniques**
- Design & build **dashboards** & other visualizations in **Tableau**
- Web interface (submit **ML/non-ML jobs** through webpage**, track**, manage)
- Automating various **data quality**, **processing** and **analysis** tasks
- Aware on **Pytorch** (data and model parallelization)
- AI Acceleration, **Data Loading** acceleration
- **Data preprocessing** & cleaning with various **Machine Learning** techniques
- Perform **EDA Task** with **Seaborn, Matplotlib**
- Perform feature engineering with **Python Libraries**
- Designing and Developing the **Machine Learning Model**.
- Analyze **Model prediction accuracy** using classification reports, confusion **matrix.** Precision, Recall, F1-Score, ROC-AUC score.
- Implementing advanced **MLOps** techniques for improving model accuracy

Project Handle:

Project Title: Automated Equipment Downtime Classification by fine-tuning DistilBERT

Description: Developed a **DistilBERT-based text classification model** to analyze and categorize **engineer downtime comments**. Extracted and preprocessed unstructured text data, manually labeled a subset for supervised learning, and fine-tuned the model to classify failure causes. Achieved **high accuracy and F1-score**, enabling real-time classification of new comments. Integrated the model into the system to support **proactive maintenance, trend analysis, and reducing invalid test occurrences**.

Technologies used: **Python, NLP, DistilBERT, Transformers, Hugging Face, Pandas, Scikit-learn, PyTorch**.

CONTACT

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EDUCATION:`

BE Mechanical (2016)
Diploma Mechanical (2013)

Achievements:

- Consistently maintained a strong performance record, contributing to key project successes.
- Awarded with **GEMS & Spot** Award for outstanding performance in Month **April**.

SKILLS:

Generative AI ,LLM Fine-tuning: Hugging Face Transformers, OpenAI API, AWS Bedrock.
RAG: Chunking, Re-ranking, retrieval pipelines.
Prompt Engineering: Few-shot, chain-of-thought (CoT), and ReAct prompting.
Fine-tuning (PEFT):LoRA, QLoRA.
Vector Databases & Embedding's: FAISS, ChromaDB, Pinecone, Weaviate.
LangChain & Llama Index: Retrievers, chains, agents, and memory.
Evaluation Matrices: BLEU, ROUGE, METEOR, BERTScore, Perplexity.
Multimodal AI: Stable Diffusion, DALL·E, CLIP, Whisper, VLMs
Responsible AI & Bias Mitigation: Bias detection, fairness, interpretability,RLHF.
Generative AI Frameworks & APIs: Hugging Face Hub,ollama,Groq.

Machine Learning:
Supervised: Linear Regression, Linear Mixed effect, Logistic Regression, Lasso and ridge regression, KNN ,Naïve Bayes, Decision Tree, Bagging / Boosting : Random forest, Gradient Boosting, AdaBoost,
Unsupervised : K-Mean Clustering, DB-Scan ,GMM, Hierarchical clustering
Python ML Packages: NumPy, SciPy Pandas, Scikit-learn, Seaborn, Matplotlib
Text Processing: NLTK, Term Frequency-Inverse Document Frequency (TF-IDF), Bag of Words, Word2vec.

Deep Learning:
ANN, **CNN,** Tensor flow, Keras, Transfer Learning Model's .

MLOPS: Docker, CICD Pipelines, Github Action,Kubernetis,MLFLOW,kubeflow, terraform ,Feature store(FEAST)
Database: **MySQL.**

Cloud Platforms: **AWS**

2) **Project Title: RAG-Based PDF Question Answering System Using AWS Bedrock & Anthropic Model**

Description: Designed and deployed a **Retrieval-Augmented Generation (RAG)** system leveraging **AWS Bedrock Anthropic models** to assist service personnel in extracting root cause insights from technical service manuals. Implemented document parsing, text chunking, and vector-based retrieval for precise question answering. **Integrated LLMs** with **ChromaDB** to enhance response accuracy and contextual understanding. Developed a **Gradio-based** front-end for seamless interaction. Optimized retrieval and generation pipelines for efficient and scalable document querying.

Technologies used: AWS Bedrock, Anthropic Models, Python, LangChain, FAISS/ChromaDB, Hugging Face, Gradio, PyTorch.

3) **Project Title: Analyzed sensor data to identify thermal fault, build and validate model for VFD thermal analytics, deployment workflow and sales package opportunities.**

Description: Approach used to detect VFD Fault is Machine learning based (Gaussian Mixed Model) clustering with Linear mixed effect (LME) Model. The resulting predictive model demonstrated remarkable accuracy in GMM diagnosing VFD fault.

4) **Project Title**: **AI-Enhanced Optimization of Psychometric Chamber Testing: Leveraged decision tree algorithms to significantly reduce invalid HVAC tests by identifying and addressing root causes, leading to more accurate and efficient system evaluations.**

Description:

Objective: Aimed to improve the reliability and efficiency of heating and cooling tests for residential air conditioners within a psychometric chamber designed for development and performance evaluation.

Root Cause Analysis: Leveraged historical test data to conduct an in-depth analysis of the factors leading to invalid tests. The goal was to uncover patterns and root causes that contributed to the high rate of test failures.

Decision Tree Implementation: Developed a robust decision tree model to systematically identify key variables responsible for invalid tests. This model provided a clear, actionable framework for predicting and preventing test failures.

Data-Driven Insights: The decision tree model revealed critical insights into the underlying causes of invalid tests, including environmental conditions, equipment calibration issues, and process anomalies, allowing for targeted interventions.

Process Optimization: Based on the findings, implemented specific changes to testing protocols and chamber conditions, significantly improving the validity and reliability of subsequent tests.

Impact: The initiative resulted in a dramatic reduction of invalid tests, from 10% to just 3%, leading to substantial cost savings, enhanced testing efficiency, and more reliable performance evaluations of residential AC systems.

Outcome: The project not only optimized resource utilization and reduced waste but also established a more rigorous and dependable testing environment, contributing to higher quality and performance standards in air conditioning system evaluations.

5) **Project Title: Anomaly Detection and Root Cause Analysis of Compressor Failures in Residential AC Systems Using Auto encoder Models during Psychometric Chamber Testing**

Description:

Anomaly Detection with Autoencoders: Developed and implemented an autoencoder neural networks-based anomaly detection model to identify early signs of compressor failure in real-time. This model was trained on historical test data to learn normal operational patterns and detect deviations indicative of potential failures.

Model Refinement and Optimization: Continuously refined the autoencoder model by incorporating feedback from detected anomalies, ensuring that the system became increasingly adept at identifying subtle failure patterns and reducing the occurrence of undetected issues.

Impact: Significantly reduced compressor-related test failures, decreasing invalid tests, saving costs, and improving the accuracy and reliability of AC performance evaluations.

Outcome: The initiative successfully integrated advanced machine learning techniques into the testing process, establishing a more reliable and efficient framework for detecting and addressing compressor failures. This contributed to higher standards of quality and performance in HVAC system development and testing.

LANGUAGES:

English / Hindi / Marathi